

**[SINGLE CORRECT CHOICE TYPE]**

**Q.1 to Q.10** has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 Let  $f(x) = \frac{e^{\ln x} \cdot \operatorname{sgn}(x^2 + x + 1)}{\ln x}$ . If  $f(x) = k$  has one distinct solution then range of  $k$  is,  
(A)  $(-\infty, 0)$  (B)  $(0, e]$  (C)  $(-\infty, 0) \cup \{e\}$  (D)  $[e, \infty)$
- Q.2 The minimum value of function  $f(x) = \frac{u^2}{x} + \frac{v^2}{u-x}$ ,  $u > 0, v > 0$  in  $(0, u)$ , is  
(A)  $\frac{1}{u}(u+v)^2$  (B)  $\frac{1}{v}(u+v)^2$  (C)  $\frac{1}{u^2}(u+v)^2$  (D)  $\frac{1}{v^2}(u+v)^2$
- Q.3 A conical cup is constructed from circular piece of paper of radius 10 cm by cutting out a sector and joining the resulting edges. If the volume of cone is maximum, then height of the cone is equal to  
(A)  $\frac{5}{\sqrt{3}}$  (B) 5 (C)  $\frac{10}{\sqrt{3}}$  (D)  $\frac{\sqrt{5}}{3}$
- Q.4 If the greatest value of  $y = \frac{x}{\ln x}$  on  $[e, e^3]$  is  $u$  then  $\frac{e^3}{u}$  is equal to  
(A) 1 (B) 2 (C) 3 (D) 4
- Q.5 The complete set of positive values of 'a' for which the equation  $|\sin 2x| - |x| - a = 0$  does not have any real solution is  
(A)  $\left(\frac{3\sqrt{3} + \pi}{6}, \infty\right)$  (B)  $\left(\frac{3\sqrt{3} - \pi}{6}, \infty\right)$  (C)  $\left(\frac{3\sqrt{3} - \pi}{12}, \infty\right)$  (D)  $\left(\frac{3\sqrt{3} + \pi}{12}, \infty\right)$
- Q.6 Let  $f(x) = \sqrt{x-1} + \sqrt{x+24-10\sqrt{x-1}}$ ,  $1 \leq x \leq 26$  be a real valued function, then for  $1 < x < 26$ ,  
(A)  $f(x) = 0$  has one real root (B)  $f'(x) = 0$  has no real root  
(C)  $f(x) = 0$  has more than one real root (D)  $f'(x) = 0$  has more than one real root
- Q.7 The number of solution(s) of the equation  $x^3 + 3x^2 + 3x - (x-6)^{1/3} + 8 = 0$  is(are)  
(A) 4 (B) 3 (C) 2 (D) 1

- Q.8 If the function  $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ , where  $a > 0$ , attains its maximum and minimum at  $p$  and  $q$  respectively such that  $p^2 = q$ , then  $a$  equals
- (A) 3 (B)  $\frac{1}{4}$  (C) 2 (D)  $\frac{1}{2}$
- Q.9 If  $f(x) = -2x^3 - 9x^2 - 12x + 6$ , then the maximum value of function  $g(x) = f(|x - 2|)$  is
- (A) 11 (B) 10 (C) 6 (D) 4
- Q.10 If the curves  $\frac{x^2}{\alpha} + \frac{y^2}{4} = 1$  and  $y^3 = 16x$  intersect at right angles, then  $\alpha =$
- (A)  $\frac{3}{4}$  (B) 2 (C)  $\frac{4}{3}$  (D)  $\frac{1}{2}$

### [PARAGRAPH TYPE]

Q.11 to Q.13 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

#### Paragraph for question no. 11 to 13

Let  $y = f(x)$  be a differentiable function passing through  $M(1, -1)$  and has a property that the chord joining any two points  $A(x_1, f(x_1))$  and  $B(x_2, f(x_2))$  always intersects y-axis at  $N(0, 2x_1x_2)$ .

- Q.11 The value of definite integral  $\int_0^{\frac{1}{2}} f(x) dx$  is equal to

(A)  $\frac{1}{6}$  (B)  $\frac{1}{8}$  (C)  $\frac{1}{12}$  (D)  $\frac{1}{24}$

- Q.12 The largest interval in which  $f(x)$  is monotonically increasing, is

(A)  $\left(-\infty, \frac{1}{2}\right]$  (B)  $\left[\frac{-1}{2}, \infty\right)$  (C)  $\left(-\infty, \frac{1}{4}\right]$  (D)  $\left[\frac{-1}{4}, \infty\right)$

- Q.13 In which of the following intervals, the Rolle's theorem is applicable to the function  $F(x) = f(x) + x$ ?
- (A)  $[-1, 0]$  (B)  $[0, 1]$  (C)  $[-1, 1]$  (D)  $[0, 2]$

### [MULTIPLE CORRECT CHOICE TYPE]

Q.14 to Q.16 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.14 Let  $g(x) = f(x) + f(1 - x)$  and  $f''(x) < 0$  in  $0 \leq x \leq 1$  then

(A)  $g(x)$  decreases in  $(0, 1)$  (B)  $g(x)$  increases in  $\left(0, \frac{1}{2}\right)$   
 (C)  $f'(x)$  is a decreasing function (D)  $f(x)$  is an increasing function

- Q.15 Consider  $f(x) = \frac{x(e^x - e^{-x}) + 2}{2x}$ ,  $x \neq 0$ . Identify which of the following statement(s) is/are correct-
- (A)  $f(x)$  has no critical points  
 (B)  $f(x)$  has exactly two critical points  
 (C)  $f(x)$  has exactly one point of inflection  
 (D)  $f(x)$  is concave downwards in  $(-\infty, 0)$  and concave upwards in  $(0, \infty)$
- Q.16 If  $f(x) = \min. \{2 \sin x, 1 - \cos x, 1\}$ , then which of the following is true.
- (A)  $\int_0^{2\pi} f(x) dx = \left( \frac{5\pi}{6} - 3 - \sqrt{3} \right)$   
 (B)  $f$  is non derivable at 2 points in  $x \in [0, 2\pi]$ .  
 (C) The number of inflexion points of  $f$  in  $[0, 2\pi]$  are 2.  
 (D) if  $g'(x) = f(x)$  in  $[0, 2\pi]$  then  $g$  is concave up in  $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$  and has a maximum at  $x = \pi$ .

### [MATRIX TYPE]

Q.17 has **three** statements (A, B, C) given in **Column-I** and **four** statements (P, Q, R, S) given in **Column-II**. Any given statement in **Column-I** can have correct matching with one or more statement(s) given in **Column-II**.

| Q.17 | Column-I                                                                                                                                                           | Column-II           |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| (A)  | The curves $y = kx^2$ and $y = e^x$ will have atleast two points in common if $k$ equals                                                                           | (P) $e^3$           |
| (B)  | If a common tangent to $y = x^2 + 1$ and $y^2 = -16(x + 1)$ touch them at $(p, q)$ and $(r, -4)$ respectively and $k \in \{p, q, r\}$ , then $e^k$ may be equal to | (Q) $e^2$           |
|      |                                                                                                                                                                    | (R) $e$             |
| (C)  | $\lambda x^2 e^x = 1$ has atleast two solutions then $\lambda$ may be equal to                                                                                     | (S) $\frac{1}{e^2}$ |

### [INTEGER TYPE]

**Q.18 to Q.20** are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

Q.18 If the function  $f(x) = x^4 + bx^2 + 8x + 1$  has a horizontal tangent and a point of inflection for the same value of  $x$ , then find the value of  $|b|$ .

Q.19 Let  $f(x) = \frac{x^2}{4}(2 \ln x - 1) - ex + k$ ,  $k \in \mathbb{R}$ . If least value of  $k$  for which  $\sqrt{f(x)}$  is defined for all  $x \in (0, \infty)$  is  $\frac{a}{b}e^2$ , where  $a, b \in \mathbb{N}$ , then find least value of  $(a + b)$ .

Q.20 Let  $f(x) = 1 + 3x^2 + 5x^4 + 7x^6 + \dots + 21 \cdot x^{20}$ ,  $x \in \mathbb{R}$   
and  $g(x) = -x^2 + 4\cos^2 \theta - 4 \sin \theta - 7$ ,  $\theta \in \mathbb{R}$   
If  $d$  is the shortest distance between  $f(x)$  &  $g(x)$  and  $d_1, d_2$  are the least and greatest value of  $d$  respectively, then find  $\left(\frac{d_2}{d_1}\right)$ .

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### ANSWER KEY

|      |      |      |                                       |      |   |      |    |      |    |
|------|------|------|---------------------------------------|------|---|------|----|------|----|
| Q.1  | C    | Q.2  | A                                     | Q.3  | C | Q.4  | C  | Q.5  | B  |
| Q.6  | D    | Q.7  | D                                     | Q.8  | C | Q.9  | C  | Q.10 | C  |
| Q.11 | D    | Q.12 | C                                     | Q.13 | B | Q.14 | BC | Q.15 | BD |
| Q.16 | ABD  | Q.17 | (A) P, Q, R; (B) Q, R, S; (C) P, Q, R |      |   |      |    |      |    |
| Q.18 | 0006 | Q.19 | 7                                     | Q.20 | 4 |      |    |      |    |

